

Functional Requirements

Functional requirements describe what the system should do.

ID	Requirement
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| FR-01 | The system shall allow patients to register and log in securely. |
| FR-02 | The system shall allow doctors to log in using their credentials. |
| FR-03 | The system shall allow patients to book appointments with doctors. |
| FR-04 | The system shall allow doctors to view their appointments. |
| FR-05 | The system shall allow doctors to add consultation notes for patients. |
| FR-06 | The system shall encrypt consultation data before storing it in the database. |
| FR-07 | The system shall allow patients to upload medical reports securely. |
| FR-08 | The system shall allow doctors to upload prescriptions. |
| FR-09 | The system shall allow patients to view and download their prescriptions. |
| FR-10 | The system shall maintain an audit log of important activities. |
| FR-11 | The system shall provide different dashboards for Admin, Doctor, and Patient. |
| FR-12 | The system shall allow only authorized users to access medical records. |

Non-Functional Requirements

These requirements describe how the system should perform.

Security

- User passwords should be stored securely using hashing.
- Consultation data should be encrypted before storage.
- Only authorized users can access medical records.

Performance

- Login should complete within a few seconds.
- Medical records should load quickly.

Reliability

- The system should save consultation data without loss.
- User data should remain consistent and secure.

Usability

- The interface should be simple and easy to use.
- Users should be able to navigate without technical knowledge.

Scalability

- The system should support adding more doctors and patients in the future.
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User Roles

Administrator

Responsibilities:

- Manage doctor accounts.
 - Manage patient accounts.
 - View system reports.
 - Monitor audit logs.
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Doctor

Responsibilities:

- Login securely.
 - View appointments.
 - Add consultation notes.
 - Upload prescriptions.
 - View patient medical history.
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Patient

Responsibilities:

- Register and login.
 - Book appointments.
 - Upload medical reports.
 - View consultation history.
 - Download prescriptions.
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System Modules

The MediCrypt system is divided into the following modules:

1. Authentication Module

- User Registration
- User Login
- Secure Authentication

2. Appointment Module

- Book Appointment
- View Appointments

3. Consultation Module

- Add Consultation Notes
- View Consultation History

4. Medical Record Module

- Upload Medical Reports
- View Medical Records

5. Prescription Module

- Upload Prescription
- View Prescription

6. Encryption Module

- Encrypt Consultation Data
- Protect Medical Records

7. Audit Log Module

- Record User Activities
 - Monitor System Access
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